Kidney Filtration and Dialysis Simulation

Quantitative Physiology | Optional practice

Ungraded • No submission required

In class, we modeled kidneys as many small filtering units. Now, extend this model to explore more realistic scenarios: how waste accumulates as kidney function declines and how intermittent dialysis or continuous machine support affects waste concentration over time.

1 Understand the model

The body continuously produces waste. Kidneys and a dialysis machine remove it. Assume one well-mixed body-fluid compartment with constant volume and constant waste production.

$$dC/dt = G/V - [(K_kidney + K_machine(t))/V]\, C$$

Symbol	Meaning	Units
C	Waste concentration	mg/L
G	Waste production rate	mg/hour
V	Waste distribution volume	L
K_kidney	Kidney clearance	L/hour
K_machine(t)	Machine clearance over time	L/hour

Clearance is the equivalent volume of fluid cleared of waste per hour. It does not mean this volume of fluid leaves the body.

2 Compare four situations

Situation	Model setup
Normal kidneys	Continuous kidney clearance; no machine.
Reduced kidney function	80% of units function. Compare no compensation, 20% extra work, and 25% extra work per remaining unit.
Intermittent dialysis	Zero kidney clearance. Machine runs Monday, Wednesday, and Friday, 08:00–12:00.
Continuous support	Zero kidney clearance. Lower machine clearance runs continuously.

In-center hemodialysis commonly runs three times weekly for about four hours. Some ICU patients receive continuous renal replacement therapy (CRRT). Zero kidney clearance is an illustrative scenario, not a condition shared by every patient receiving these treatments. [1–2]

3 Set parameters and compensation

Use the following illustrative values for this educational simulation only.

Parameter	Value
Fluid volume V	40 L
Waste production G	100 mg/hour
Baseline kidney clearance K_o	4 L/hour
Initial concentration C(0)	25 mg/L
Intermittent machine clearance when running	12 L/hour
Continuous machine clearance	2 L/hour
Simulation duration	14 days, starting Monday at 00:00

$$K_kidney = K_0 f \alpha$$

Here f is the fraction of functioning filtering units and α is their workload multiplier. With 20% of units lost, 20% extra work gives 0.8 × 1.2 = 0.96, or 96% of baseline clearance. Maintaining 100% requires α = 1/0.8 = 1.25, or 25% extra work. This assumed compensation is not a guaranteed biological response.

4 Explore and explain

- Plot all scenarios together and shade intermittent treatment periods. Why does waste rise between treatments? What happens during the Friday-to-Monday gap?
- Compare the highest, lowest, and time-averaged concentrations during week 2. Does 20% extra work fully compensate for losing 20% of filtering units?
- Skip the first Friday session. Separately, pause continuous support for six hours. Explain the changes in the curves.

Additional comparison: the original machine settings provide different integrated weekly clearance. Match this quantity by changing continuous clearance to:

$$K_continuous = (12\ L/hour \times 12\ hours/week)/(168\ hours/week) \approx 0.857\ L/hour$$

Do equal integrated weekly clearances produce identical peaks, fluctuations, or average concentrations? Explain using your plots.

5 Validate the simulation

With no clearance, verify C(t) = C(0) + (G/V)t. With constant positive total clearance K, verify that concentration approaches Csteady = G/K. Concentration must stay nonnegative and continuous at machine switching times. Repeat with a smaller time step to check convergence.

Limitations: This model tracks one waste substance in one compartment. It omits fluid balance, electrolytes, blood pressure, exchange between body compartments, and hormonal kidney functions. It cannot determine treatment needs.

6 Prompt for Colab Gemini

Paste the following prompt into Gemini in Google Colab. Read and explain the generated code before modifying it.

```
Create a beginner-friendly Python simulation of waste clearance by kidneys and dialysis
using:
dC/dt = G/V - (K_kidney + K_machine(t))*C/V.

Use V=40 L, G=100 mg/hour, C0=25 mg/L, and baseline kidney clearance K0=4 L/hour. Simulate
14 days starting Monday at 00:00.

Compare:
1. Normal kidneys: K_kidney=4, no machine.
2. Reduced kidneys: K_kidney=4*0.8*alpha, with alpha=1, 1.2, 1.25.
3. Zero kidney clearance with intermittent dialysis: K_machine=12 L/hour,
Monday/Wednesday/Friday from 08:00–12:00 each week; zero at other times.
4. Zero kidney clearance with continuous support: K_machine=2 L/hour.

Plot waste concentration versus time in days. Shade dialysis sessions. Also plot the machine
on/off schedule. Label all units.

Use clearly commented code and explain each model term. Handle treatment start and stop
times exactly so the numerical solver does not skip sessions.

Add optional comparisons:
- Skip the first Friday session.
- Pause continuous support for six hours, starting on day 3 at 08:00. Label this
interruption clearly.
- Set continuous clearance to 12*12/168 L/hour to match intermittent integrated weekly
clearance.

Report minimum, maximum, and time-averaged concentrations during week 2. Validate against
the analytic solutions for zero clearance and constant clearance, and check numerical
convergence.

Label all parameters as illustrative. Do not add clinical danger thresholds or treatment
recommendations.
```

7 Optional challenge

A week in a patient's life

Quantitative Physiology • Kidney Filtration and Dialysis Simulation

Choose one scenario to explore using the kidney filtration model from the optional practice. Use illustrative assumptions for this educational simulation only.

Scenario A A missed dialysis appointment

A person normally receives dialysis on Monday, Wednesday, and Friday but misses Wednesday because transportation is unavailable. Compare attending the next scheduled session on Friday with receiving a rescheduled session on Thursday. How do the peak waste concentration and time spent above the usual weekly range differ?

Scenario B Meals change waste production

Waste production varies throughout the day. Model a temporary increase in the production of a urea-like waste after meals. How do meal timing and dialysis timing affect the concentration curve?

Scenario C A temporary interruption in the ICU

A patient receiving continuous machine support has a four-hour interruption for a procedure or filter replacement. Assume waste production continues during this period. How much does waste concentration rise, and how long after restarting does it return close to its previous level?

For your chosen scenario

State your assumptions, compare the scenario with uninterrupted scheduled treatment on one graph, and explain the results and one model limitation. These comparisons explore model behavior rather than determine treatment decisions.

Optional; no separate submission required.

8 Optional challenge

Improving dialysis access with ML and optimization

Quantitative Physiology: How can we coordinate appointments, transportation, and available machines so that more patients can attend their prescribed treatment?

In this fictional scenario, three dialysis centers serve several communities. Patients differ in their travel distance, transportation options, and availability. Each center has a limited number of staffed treatment slots. Your task is to combine machine learning, scheduling optimization, and your kidney filtration model to explore improvements.

(Use synthetic data and illustrative assumptions for this educational simulation only.)

Step 1 Create your simulated community

Create a dataset containing:

- Patients: location, transportation options, availability, prescribed treatment schedule, and remaining kidney clearance.
- Centers: location, staffed machine capacity, operating hours, and appointment slots.
- Past appointments: travel time, appointment time, transportation reliability, and whether the appointment was attended.

State how you generate these synthetic records. Keep prescribed treatment frequency, duration, and allowable spacing fixed throughout the exercise.

Step 2 Use ML to predict attendance barriers

Train a decision tree or logistic regression model to estimate the probability of a missed appointment using travel time, transportation reliability, and scheduling conflicts.

Evaluate the model on held-out patients. Compare it with a simple baseline, such as predicting the same missed-appointment probability for everyone.

Question: Which factors help identify appointments that may require transportation or scheduling support?

Use predictions to offer support, not to exclude patients or reduce their access.

Step 3 Optimize appointment assignments

Use the predicted attendance probabilities to help assign patients to centers and appointment times.

Objective: Reduce expected missed appointments and travel burden while meeting prescribed treatment needs.

Your schedule must respect:

- Staffed machine capacity and operating hours.
- Patient availability and transportation schedules.
- Prescribed session duration, frequency, and spacing.
- A fixed budget for additional support.

Start with a simple scheduling rule. For a greater challenge, use an optimization solver. If no feasible schedule exists, report the unmet needs.

Step 4 Compare resource choices

Compare your optimized schedule with a baseline that assigns patients to the nearest center with an available slot. Then test one additional resource:

- Transportation support.
- An additional staffed treatment shift.
- An additional machine with the staff and operating time needed to use it.

Use a stated budget and explain why your chosen resource addresses the main barrier. Adding a machine may have little effect if patients cannot reach the center.

Step 5 Evaluate the physiological consequences

Feed the simulated attendance schedules into your kidney filtration model. Compare waste concentration when prescribed sessions are attended and when sessions are missed or delayed.

Examine whether patients with different remaining kidney clearance respond differently to the same disruption. Use these results to explain model behavior, not to set treatment requirements.

What to compare

- Expected and simulated missed appointments.
- Travel time and unmet treatment needs.
- Waste concentration over time.
- Access for remote communities compared with communities near a center.

Repeat the simulation with different random outcomes. Explain whether the improvement persists and how much it depends on your assumptions. Changing an appointment changes its travel and scheduling features, so update its predicted attendance probability.

What to present

Provide an appointment chart or map, a before-and-after comparison, and a brief explanation of your assumptions and limitations.

ML predicts attendance barriers; optimization proposes schedules; physiology simulation evaluates waste accumulation.

Optional; no separate submission required.

Background reading

1. NIDDK. Choosing a Treatment for Kidney Failure. https://www.niddk.nih.gov/health-information/kidney-disease/kidney-failure/choosing-treatment

2. Nationwide Children's Hospital. Continuous Renal Replacement Therapy. https://www.nationwidechildrens.org/family-resources-education/health-wellness-and-safety-resources/helping-hands/continuous-renal-replacement-therapy